Exercise

Consider the following stochastic process:

$$y(t) = \frac{1}{6}\, y(t-1) + \frac{1}{6}\, y(t-2) + \eta(t) \qquad \eta \sim WN(0, 3)$$

⇒ Write the recursive equation of the optimal 1-step ahead predictor

1. Compute the T.F.
2. Check if it is written in canonical form
3. Apply the predictor formula

DATA

$$\hat{y}(t+1|t) = \frac{C(z) - A(z)}{C(z)} \cdot y(t+1)$$

1. $$W(z) = \frac{1}{1 - \frac{1}{6} z^{-1} - \frac{1}{6} z^{-2}} = \frac{z^2}{z^2 - \frac{1}{6} z - \frac{1}{6}}$$

- Zeros: $z_{1,2} = 0$
- Poles: $z_{1,2} = \dfrac{\frac{1}{6} \pm \sqrt{\frac{1}{36} + \frac{4}{6}}}{2}$ $=$ $z_1 = \frac{1}{2}$, $z_2 = -\frac{1}{3}$

Both are inside the unitary circle
⇓
Process y is stationary

2.
- Same order ✓
- Monic ✓
- Coprime ✓
- Roots inside unit circle ✓

$$\Rightarrow W(z) = \frac{C(z)}{A(z)}$$

3. Compute predictor:

$$\hat{y}(t+1|t) = \frac{\cancel{1} - \cancel{1} + \frac{1}{6} z^{-1} + \frac{1}{6} z^{-2}}{1} \cdot y(t+1)$$

$$= \left[\frac{1}{6} + \frac{1}{6} z^{-1}\right] \cdot z^{-1} \cdot y(t+1)$$

$$= \left[\frac{1}{6} + \frac{1}{6} z^{-1}\right] \cdot y(t)$$

$$\hat{y}(t|t-1) = \left[\frac{1}{6} + \frac{1}{6} z^{-1}\right] \cdot y(t-1)$$

⇓

$$\boxed{\hat{y}(t+1|t) = \frac{1}{6}\, y(t) + \frac{1}{6}\, y(t-1)}$$

Recursive equation

⇒ Compute the prediction error variance

Theory lectures

$$\mathrm{Var}\left[\varepsilon(t+1|t)\right] = \mathrm{Var}\left[\hat{y}(t+1|t) - y(t+1)\right] = \mathrm{Var}\left[\eta(t)\right] = \boxed{3}$$

⇒ WRITE THE RECURSIVE EQUATION OF THE 2-STEPS AHEAD PREDICTOR

WE NEED TO WRITE THE PROCESS AS FOLLOWS:

$$W(z) = E(z) + \frac{R(z)}{A(z)} \sim f(z^{-2}, \dots) \Rightarrow \hat{y}(t+2|t) = \frac{R(z)}{C(z)} \cdot y(t+2)$$

$E(z)$: $f(\dots, z^{-1})$

TO COMPUTE $R(z)$ AND $E(z)$ WE APPLY THE LONG DIVISION:

$$\begin{array}{l|l} 1 & 1 - \tfrac{1}{6} z^{-1} - \tfrac{1}{6} z^{-2} \\ \underline{1 - \tfrac{1}{6} z^{-1} - \tfrac{1}{6} z^{-2}} & \boxed{1 + \tfrac{1}{6} z^{-1}} \\ \tfrac{1}{6} z^{-1} + \tfrac{1}{6} z^{-2} & \\ \underline{\tfrac{1}{6} z^{-1} - \tfrac{1}{36} z^{-2} - \tfrac{1}{36} z^{-3}} & \\ \boxed{\tfrac{7}{36} z^{-2} + \tfrac{1}{36} z^{-3}} & \end{array}$$

$E(z) = f(\dots z^{-1})$

$R(z) = f(z^{-2}, \dots)$

SO THE PREDICTOR IS:

$$\hat{y}(t+2|t) = \frac{\left[\frac{7}{36} + \frac{1}{36} z^{-1}\right]}{1} \cdot z^{-2} \cdot y(t+2)$$

$$= \left[\frac{7}{36} + \frac{1}{36} z^{-1}\right] \cdot y(t)$$

⇓

$$\boxed{\hat{y}(t+2|t) = \frac{7}{36} \cdot y(t) + \frac{1}{36} y(t-1)}$$

⇒ COMPUTE THE PREDICTION ERROR VARIANCE

THEORETICAL CLASS

$$\mathrm{Var}\left[\varepsilon(t+2|t)\right] = \mathrm{Var}\left[\hat{y}(t+2|t) - y(t+2)\right] =$$

THE PREDICTION ERROR IS A STOCHASTIC PROCESS GENERATED BY FEEDING $E(z)$ WITH $\eta(t)$

SINCE $E(z)$ IS A MA(1) PROCESS THE VARIANCE IS:

$$\mathrm{Var}\left[\varepsilon(t+2|t)\right] = \gamma_\varepsilon(0) = (c_0^2 + c_1^2) \cdot \lambda_\eta^2$$

$$= \left[1 + \frac{1}{36}\right] \cdot 3 = \boxed{\frac{37}{36} \cdot 3}$$

CONSISTENTLY WE GET THAT:

$$\mathrm{Var}\left[\, \varepsilon(t+2|t) \,\right] > \mathrm{Var}\left[\, \varepsilon(t+1|t) \,\right]$$

Exercise

Consider the following stochastic process:

$$y(t) = \frac{1}{3}\,y(t-1) + u(t-2) + 3e(t-1) + e(t-2) \qquad e(t) \sim WN(0,1)$$

⇒ Classify the process

The process is $ARMAX(1,2,2)$

To check if the process is stationary we have to isolate the deterministic and the stochastic part:

$$y(t) = \frac{1}{3}\cdot z^{-1}\cdot y(t) + u(t)\, z^{-2} + \left[3\cdot z^{-1} + z^{-2}\right] e(t)$$

$$y(t) = \underbrace{\frac{z^{-2}}{1-\frac{1}{3}z^{-1}}\cdot u(t)}_{\text{deterministic part}} + \underbrace{\frac{3z^{-1}+z^{-2}}{1-\frac{1}{3}z^{-1}}\cdot e(t)}_{\text{stochastic part}}$$

Poles of the stochastic part: $z_1 = 1/3$ ⇒ process is stationary

⇒ Find the minimum order representation of the stochastic part of the system

- same order ✗
- monic ✗
- coprime
- roots inside unit circle ✓ ✓

$$v(t) = \frac{3\cdot z^{-1}+z^{-2}}{1-\frac{1}{3}z^{-1}}\cdot e(t) = \frac{3+z^{-1}}{1-\frac{1}{3}z^{-1}}\cdot \boxed{z^{-1}\cdot e(t)} = \frac{1+\frac{1}{3}z^{-1}}{1-\frac{1}{3}z^{-1}}\cdot \boxed{3\cdot\tilde{e}(t)}$$

$\tilde{e}(t) \sim WN(0,1)$

$\eta(t) \sim WN(0, 1\cdot 3^2) \sim 9$

$$v(t) = \frac{1+\frac{1}{3}z^{-1}}{1-\frac{1}{3}z^{-2}}\cdot \eta(t) \qquad \eta(t) \sim WN(0,9)$$

⇒ Write the recursive equation of the optimal 2-steps ahead predictor

Write the process:

$$y(t) = \frac{B(z)}{A(z)}\cdot u(t-d) + \frac{C(z)}{A(z)}\cdot e(t)$$

$$\downarrow$$

$$\hat{y}(t+k\,|\,t) = \frac{B(z)\cdot E(z)}{C(z)}\cdot u(t-d+k) + \frac{R(z)}{C(z)}\cdot y(t+k)$$

THE REQUIRED POLYNOMIALS ARE:

- $B(z) = 1$
- $C(z) = 1 + \frac{1}{3} z^{-1}$

$$
\begin{array}{l|l}
1 + \frac{1}{3} z^{-1} & 1 - \frac{1}{3} z^{-1} \\
\cline{2-2}
1 - \frac{1}{3} z^{-1} & \boxed{1 + \frac{2}{3} z^{-1}} \leftarrow E(z) \\
\cline{1-1}
\frac{2}{3} z^{-1} & \\
\frac{2}{3} z^{-1} - \frac{2}{9} z^{-2} & \\
\cline{1-1}
\boxed{\frac{2}{9} z^{-2}} \leftarrow R(z) &
\end{array}
$$

SO THE PREDICTOR IS:

$$
\hat{y}(t+2|t) = \frac{1 + \frac{2}{3} z^{-1}}{1 + \frac{1}{3} z^{-1}} \cdot u(t-2+2) + \frac{\frac{2}{9} z^{-2}}{1 + \frac{1}{3} z^{-1}} \cdot y(t+2)
$$

$$
= \frac{1 + \frac{2}{3} z^{-1}}{1 + \frac{1}{3} z^{-1}} \cdot u(t) + \frac{\frac{2}{9}}{1 + \frac{1}{3} z^{-1}} \cdot y(t)
$$

THE RECURSIVE EQUATION IS THE FOLLOWING:

$$
\boxed{\hat{y}(t+2|t) = -\frac{1}{3} \cdot \underbrace{\hat{y}(t+1|t-1)}_{\hat{y}(t+2|t) \cdot z^{-1}} + u(t) + \frac{2}{3} u(t-1) + \frac{2}{9} \cdot y(t)}
$$

⇒ COMPUTE THE PREDICTION ERROR VARIANCE

THE PREDICTION ERROR IS ONLY DUE TO THE STOCHASTIC PART:

$$
Var[\varepsilon(t+2|t)] = \gamma_{\varepsilon}(0) = \left[1^2 + \frac{4}{9}\right] \cdot 9 = \boxed{13}
$$

Exercise

Consider the following SP

$$y(t) = \eta(t) + \tfrac{1}{3}\,\eta(t-1) \qquad \eta \sim WN(2,4)$$

⇒ **Write the equation of the optimal 1-step ahead predictor**

$$\hat{y}(t+1|t) = \frac{C(z)-A(z)}{C(z)} \cdot y(t+1) \quad *$$

The formula holds only for zero-mean S.P.

$\boxed{\text{Idea \# 1}}$

1. De-polarize the process $\quad \tilde{y}(t) = y(t) - \mu_y$, $\quad \tilde{\eta}(t) = \eta(t) - \mu_\eta$
2. Apply the predictor formula *
3. Re-polarize the process

$$\hat{y}(t+1|t) = \hat{\tilde{y}}(t+1|t) + \mu_y$$

$\boxed{1.}$

$$\mu_y = W(z=1)\cdot \mu_\eta = \left[1 + \tfrac{1}{3}\right]\cdot 2 = \tfrac{8}{3}$$

$$W(z) = \left[1 + \tfrac{1}{3} z^{-1}\right]$$

$\boxed{2.}$

$$\hat{\tilde{y}}(t+1|t) = \frac{\tfrac{1}{3} z^{-1}}{1 + \tfrac{1}{3} z^{-1}} \cdot \tilde{y}(t+1) = \frac{\tfrac{1}{3}}{1 + \tfrac{1}{3} z^{-1}} \cdot \tilde{y}(t)$$

Predictor for the de-biased process

$\boxed{3.}$

$$\hat{y}(t+1|t) = \frac{\tfrac{1}{3}}{1 + \tfrac{1}{3} z^{-1}} \cdot \tilde{y}(t) + \mu_y$$

$$= \frac{\tfrac{1}{3}}{1 + \tfrac{1}{3} z^{-1}} \cdot \left(y(t) - \mu_y\right) + \mu_y$$

$$= \frac{\tfrac{1}{3}}{1 + \tfrac{1}{3} z^{-1}} \cdot y(t) + \left[1 - \frac{\tfrac{1}{3}}{1 + \tfrac{1}{3} z^{-1}}\right]\mu_y$$

$$\hat{y}(t+1|t) = \frac{\tfrac{1}{3}}{1 + \tfrac{1}{3} z^{-1}}\, y(t) + \underbrace{\frac{1}{\tfrac{4}{3}} \cdot \tfrac{8}{3}}_{=2}$$

* The transients are neglected and the steady-state solution is used

THE RECURSIVE EQUATION OF THE PREDICTOR IS:

$$\hat{y}(t+1|t) = -1/3\,\hat{y}(t|t-1) + 1/3\,y(t) + (1 + 1/3\,z^{-1})\cdot 2$$

$$= -1/3\,\hat{y}(t|t-1) + 1/3\,y(t) + [1 + 1/3]\,2$$

$$\boxed{\hat{y}(t+1|t) = -1/3\,\hat{y}(t|t-1) + 1/3\,y(t) + 8/3}$$

RECURSIVE EQUATION

IDEA # 2

CONSIDER THE AVERAGE VALUE OF THE NOISE AS A DETERMINISTIC INPUT

$$y(t) = \eta(t) + 1/3\,\eta(t-1) \qquad \eta \sim WN(2,4)$$

$$\Downarrow \qquad \eta(t) = \delta(t) + 2 \qquad \delta \sim WN(0,4)$$

$$y(t) = (\delta(t)+2) + 1/3\,(\delta(t-1)+2)$$

$$\boxed{y(t) = \delta(t) + 1/3\,\delta(t-1) + 8/3}$$

INPUT "u"

$$\delta \sim WN(0,4)$$

EQUIVALENT MAX PROCESS!

WE CAN COMPUTE THE PREDICTOR USING THE "NORMAL" FORMULAS

$$y(t) = \frac{\boxed{1}_{\,B(z)}}{\boxed{1}_{\,A(z)}}\cdot u(t-1) + \frac{\boxed{1 + 1/3\,z^{-1}}_{\,C(z)}}{\boxed{1}_{\,A(z)}}$$

SO THE 1-STEP AHEAD PREDICTOR IS:

$$\hat{y}(t+1|t) = \frac{\boxed{1}_{\,B}\cdot\boxed{1}_{\,\tilde{E}}}{1+1/3\,z^{-1}}\cdot u(t-1+1) + \frac{1/3\,z^{-1}}{1+1/3\,z^{-1}}\cdot y(t+1)$$

$$= \frac{1}{1+1/3\,z^{-1}}\cdot u(t) + \frac{1/3}{1+1/3\,z^{-1}}\cdot y(t)$$

$$= \frac{1}{1+1/3\,z^{-1}}\cdot 8/3 + \frac{1/3}{1+1/3\,z^{-1}}\,y(t)$$

$$= \frac{1}{4/3}\cdot 8/3 + \frac{1/3}{1+1/3\,z^{-1}}\,y(t)$$

$$\boxed{\hat{y}(t+1|t) = \frac{1/3}{1 + 1/3\, z^{-1}}\, y(t) + 2}$$

SAME RESULT !